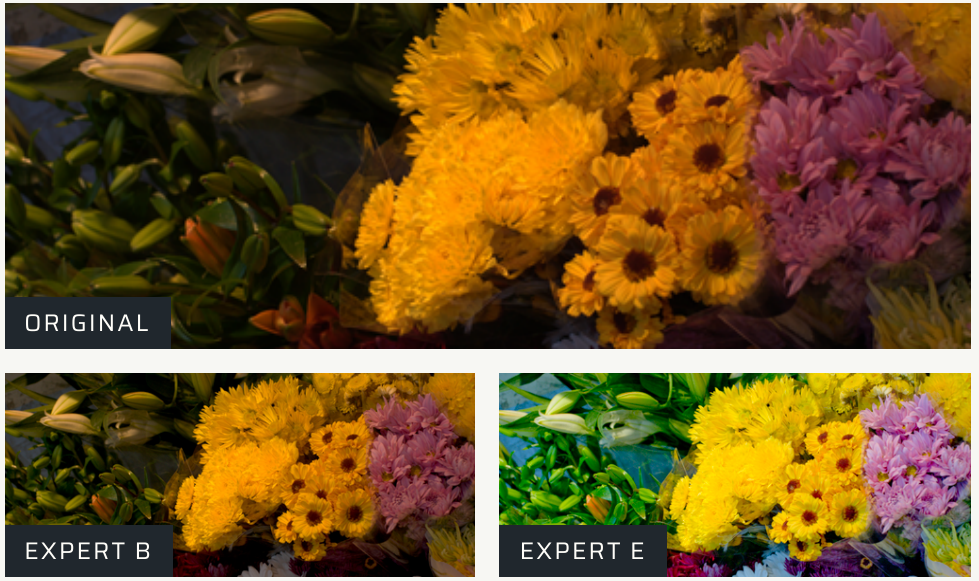
Personalized Agent-based Image Enhancement via User Preference Modeling

Learning how a user edits,
from their editing history.

```
p(z | image, history)
```



One scene, different recorded edits.
FiveK references, not model predictions.

Client Marvin Tao

Tutor Chuanzhi Xu

SOFT3888_TH18_04

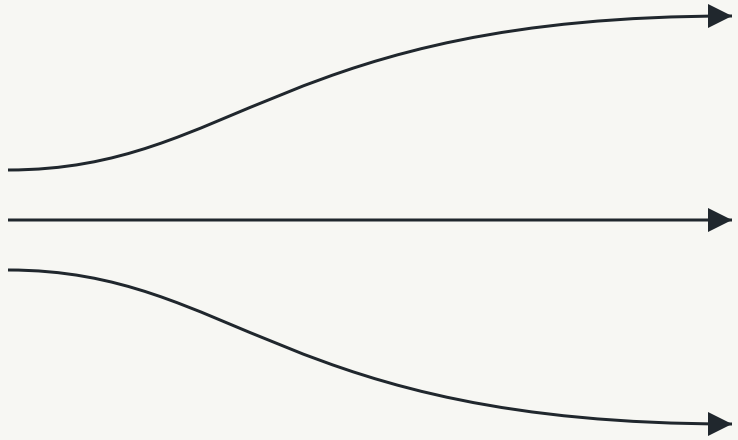
Yuchen Deng
Gary Xu
Sihan Fang
Yiming Chen
Yiheng Guo
Lunyan Wang
Weishao Su
XingTang Chen
Zihao Zhang

// 02 Project background

One image. Three users. Three preferences.



Original image



User A



User B



User C



User A Historical examples



Original



User A edit



Original



User A edit

User B Historical examples



Original



User B edit



Original



User B edit

User C Historical examples



Original



User C edit



Original



User C edit

Learn from historical edits. Adapt to unseen users without additional training. Preference illustration.

// 03 Existing methods and our core idea

01 User-specific training

AdaInt / RSFNet

A separate user or expert model.

02 Additional instructions

DiffRetouch / PerTouch

Explicit editing guidance.

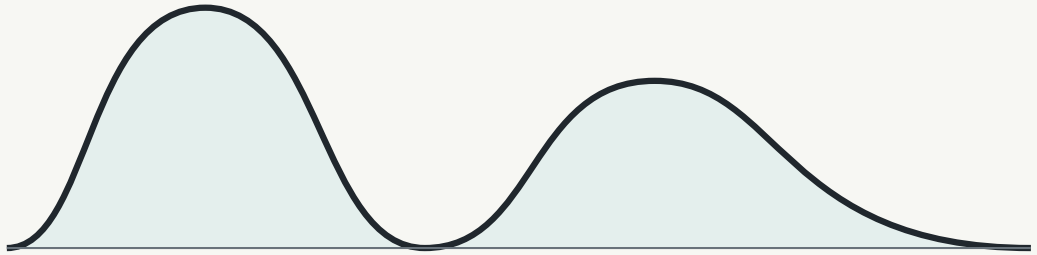
03 One preference vector

PieNet

History compressed into one representation.

OUR CORE IDEA

Learn a distribution of possible edits.



Retouching space · conceptual

`P(z | current image, user history)`

Hypothesis: one image and user may admit several acceptable edits.

// 04 Project goals and achievements

Three objectives. One shared model.

OBJECTIVE 01

Automatic

No editing instructions

3D space

Compact retouching representation

OBJECTIVE 02

Generalizable

No user-specific fine-tuning

Transformer

Image and history conditioning

OBJECTIVE 03

Multimodal

Several candidate edits

Flow

Continuous conditional density

3D edit space → Transformer condition → Conditional Flow

A two-stage architecture supports automatic, training-free, multimodal retouching.

// 05 User stories and functional requirements

Three user needs guide the system design.

FR–01

Automatic personalization

Current image + user history

↓

No editing instructions

FR–02

Adapt to new users

Eight historical examples

↓

No additional training

FR–03

Generate alternatives

Continuous preference distribution

↓

Multiple retouching solutions

Automatic editing → training-free adaptation → multimodal prediction

// 06 Acceptance criteria and constraints

REQUIREMENT	ACCEPTANCE CHECK	EVALUATION EVIDENCE
FR-01	No instructions or query targets	FiveK / PPSD · PSNR and SSIM
FR-02	8 history pairs · fixed weights	Expert E + 50 unseen PPSD users
FR-03	Multiple sampled edit candidates	Density analysis + Top-1 / Best-of-K

Constraints

- Compute budget
- History availability
- Leakage prevention
- Fixed model weights

> The query target enters scoring only after prediction.

Modelling user preferences

≈30%

Early similarity-retrieval accuracy

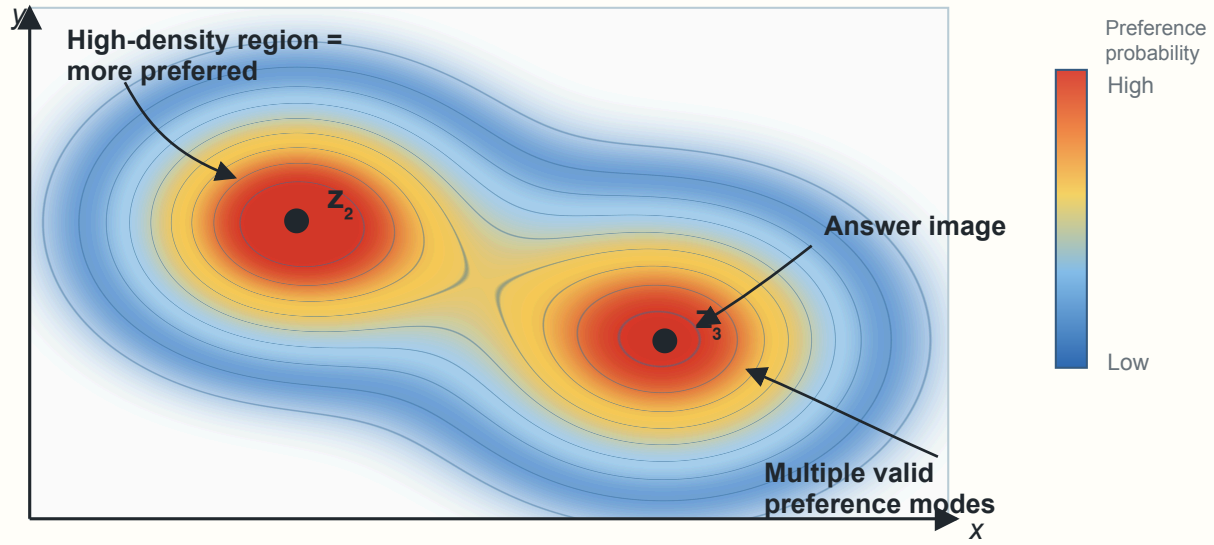
Similarity alone is not enough.

`P(z | image, history)`

A continuous retouching space allows more than one plausible edit to coexist.

Proposed Hypothesis: Preference is a Continuous Probability Density

User preference distribution in retouching space (2D example)



The answer image is only one sample located in a high-density region;
other nearby or separate high-density regions may also represent valid user preferences.

READ THE MAP

- **High density** — More plausible retouching region.
- **Low density** — Less likely under the learned distribution.

`z = (x, y)`

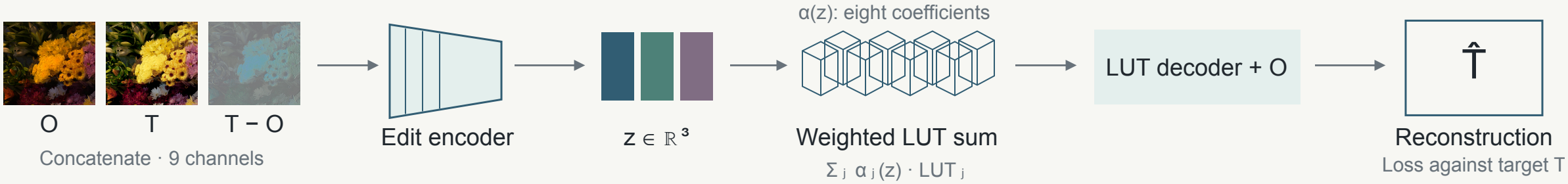
Each point is a possible retouching solution.

One recorded edit is an observation; a conditional density can represent several possible edits.

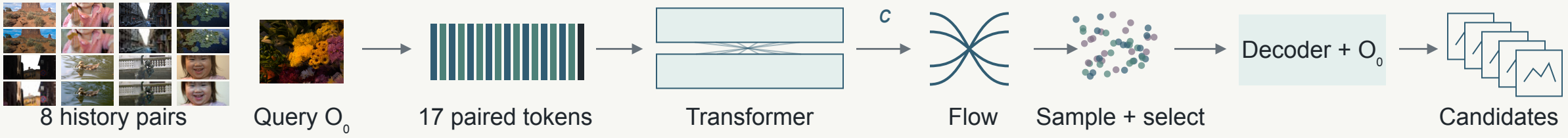
Conceptual density. A single recorded target cannot validate every candidate preference.

// 08 Overall system architecture

Stage 1 Learn the edit space



Stage 2 Predict from image + history



FiveK inputs / report architecture / illustrative samples

No query target T_0 at inference

Stage 1: compact edit vectors. Stage 2: image + history → conditional density → candidate edits.

// 09 Stage 1 · Retouching encoder and decoder



Original O



Target T

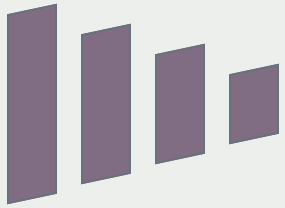


Difference
T − O

9 input channels

→

Encoder



Convolution
+ downsampling

Flatten → FC

z 3D edit vector

→

Decoder

$z \rightarrow FC \rightarrow w_1, \ldots, w_8$



LUT 1 LUT 2 LUT 3 LUT 4



LUT 5 LUT 6 LUT 7 LUT 8

$\Sigma\ w_i \cdot LUT_i$

8 learnable basis LUTs

→

Reconstruct

Apply final LUT to O



↓

$\hat{T}$

Reconstruction loss

$L_{rec}(\hat{T}, T)$ · prediction vs expert target

→

Backpropagation ↶

Jointly update encoder, decoder and basis LUTs.

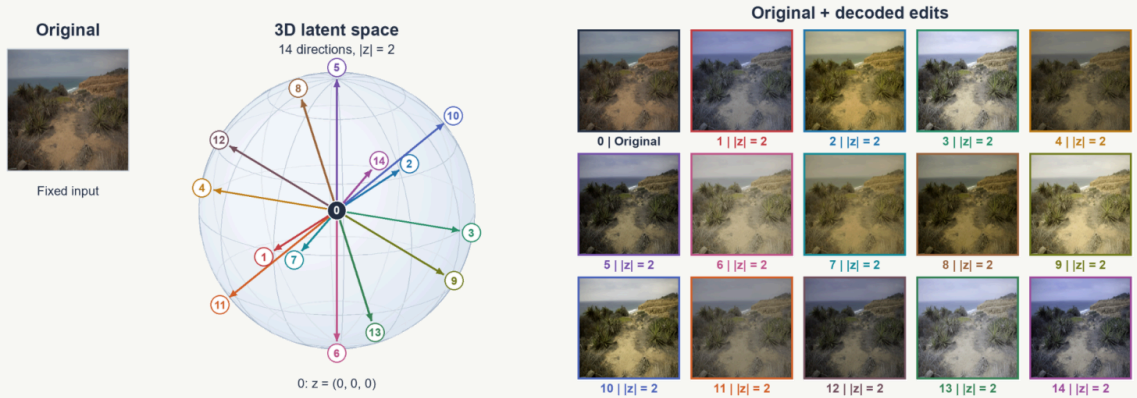
// 10 Understanding the retouching space

Continuous Nearby vectors, similar edits

$$z = \|z\| \cdot \hat{z}$$

Direction → editing type

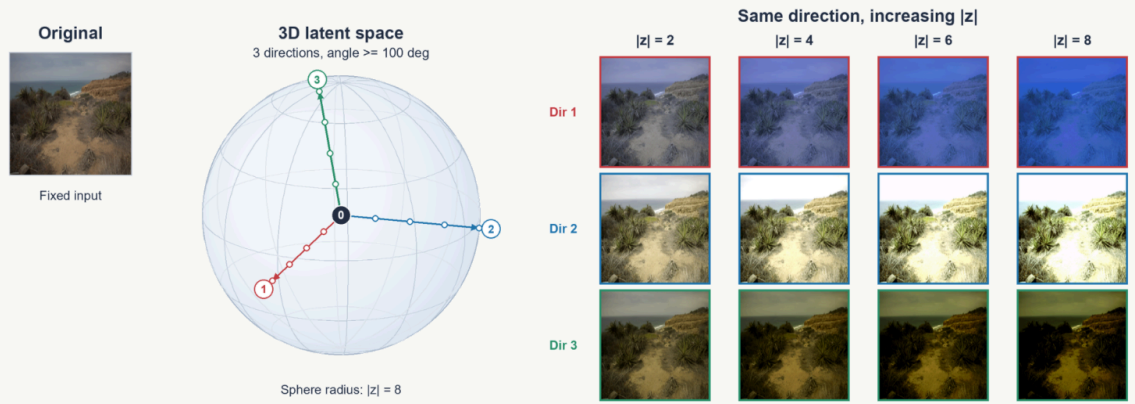
14 directions · fixed magnitude ‖z‖ = 2



Different directions change colour and tone.

Magnitude → editing strength

3 directions · magnitudes 2, 4, 6, 8



The same direction produces progressively stronger edits.

Editing type and strength vary within one continuous 3D space.

// 11 Stage 2 · Transformer-based conditioning

01 Image content

One shared encoder



O_0 · query



$O_1 \ldots O_8$

→



Shared image encoder

Conv + downsample → E_i

→

FC

$\tilde{E}_i \in \mathbb{R}^D$

02 Editing history

8 original–target pairs



O_i



T_i

→



Stage 1 edit encoder

$(O_i, T_i, T_i - O_i) \rightarrow z_i \in \mathbb{R}^3$

→

FC

$\tilde{z}_i \in \mathbb{R}^D$

Pair the history, then append the query

17 tokens · common dimension D

$\tilde{E}1$	$\tilde{z}1$	$\tilde{E}2$	$\tilde{z}2$	$\tilde{E}3$	$\tilde{z}3$	$\tilde{E}4$	$\tilde{z}4$	$\tilde{E}5$	$\tilde{z}5$	$\tilde{E}6$	$\tilde{z}6$	$\tilde{E}7$	$\tilde{z}7$	$\tilde{E}8$	$\tilde{z}8$	$\tilde{E}_0$
H1		H2		H3		H4		H5		H6		H7		H8		QUERY

FC projections align image features and 3D edit vectors to D dimensions.

↓

Two-layer Transformer

Image–edit relationships via self-attention

Layer 1

Self-attention + FFN

→

Layer 2

Self-attention + FFN

→

c

Read updated query token

$\tilde{E}'_0 = c \in \mathbb{R}^D$

To the Conditional Flow

- Image + history inform c
- Unseen users · no weight updates
- No query target at inference

// 12 Conditional Flow and candidate generation

C Current image + 8 history pairs → Transformer — Shared condition · z, u ∈ $\mathbb{R}^3$

01 / TRAINING Train the density

Maximise log p(z | c)

A
Ground-truth z



Stage 1 encodes (O, T)

→

B
Conditional Flow

f

u = f(z; c)

Invertible transformation

→

C
Standard Gaussian



u ~ N(0, I)

OPTIMISE THE MAP

Conditional log-likelihood **L = −log p(z | c)**

$$\log p(z \mid c) = \log p(u) + \log |\det J|$$

$J = \partial f(z; c) / \partial z$

$\log p(u) = -\tfrac{1}{2}\|u\|^2 - \tfrac{3}{2}\log(2\pi)$

↰ Backpropagation updates Flow parameters.

02 / INFERENCE Generate candidates

Reverse the map, then select

01
Monte Carlo



M samples u ~ N(0, I)

→

02
Inverse Flow + c

f^{-1}

$z = f^{-1}(u; c)$

M candidate edit vectors

→

03
K-means in z space



K groups of edit vectors

→

04
Rank clusters

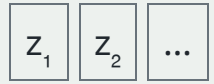


Mean log p(z | c)

One representative per cluster

→

05
Top-1 … Top-K



LUT decoder + original O

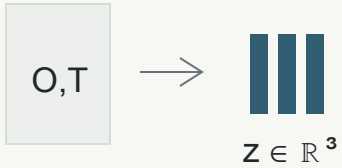
Multiple retouched images

Density-ranked alternatives No query target during generation. Best-of-K uses the target only for offline evaluation.

// 13 Testing strategy and quality assurance

01

Can the edit be represented?



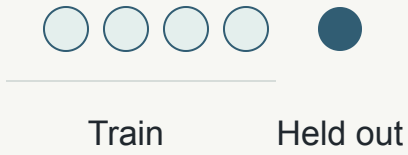
Encoder–decoder reconstruction

Test 3D representation capacity.

PSNR · SSIM · latent continuity

02

Does prediction generalise?



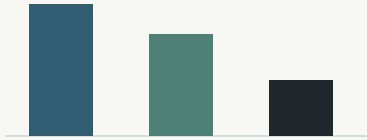
FiveK and PPSD evaluation

Test with fixed model weights.

Fidelity · held-out expert / users

03

What drives the prediction?



Input ablations and Top-K coverage

Swap conditions; compare Top-1 / Best-of-K.

PSNR changes · candidate coverage

04

What does inference cost?



Controlled GPU benchmark

Measure candidate generation.

Latency · throughput · memory

Keep prediction separate from scoring

Reconstruction, accuracy, unseen users, ablations, multimodality and computational performance.

Report §3.1 / schematic diagrams, not measured results.

// 14 Datasets and evaluation protocol

FiveK · 5,000 originals

Experts A–D · train	Expert E unseen

Five expert-retouched versions per image

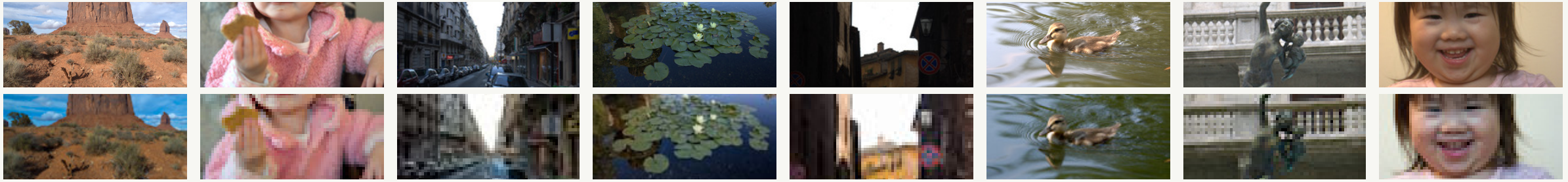
PPSD

250 train users	50 unseen

Training-free adaptation to held-out users

Eight historical pairs per query

Original + retouched image



H1 · C | H2 · C | H3 · C | H4 · C | H5 · C | H6 · C | H7 · C | H8 · C

Fixed model parameters. No query ground truth at inference.

// 15 Test cases and requirement coverage

9 controlled checks **3** requirements **Fixed** inference weights

FR–01

Automatic retouching

- TC–01 PSNR / SSIM
- TC–02 Existing methods
- TC–05 Wrong history
- TC–06 Wrong image

FR–02

Unseen users

- TC–03 Held-out Expert E
- TC–04 50 PPSD users

FR–03

Multiple candidates

- TC–07 History-conditioned density
- TC–08 Top-1 / Best-of-K
- TC–09 Candidate comparisons

Prediction → held-out evaluation → conditioning and candidate analysis

Nine experimental checks evaluate automatic retouching, unseen-user adaptation and multimodal prediction.

// 16 Retouching space reconstruction evaluation

28.5 dB

FiveK reconstruction PSNR

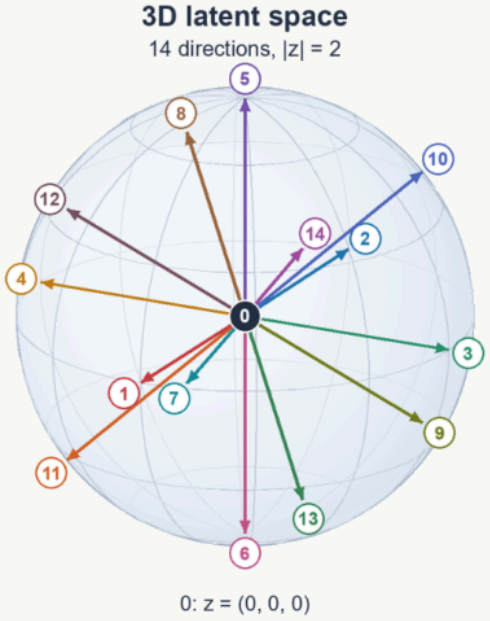
28.2 dB

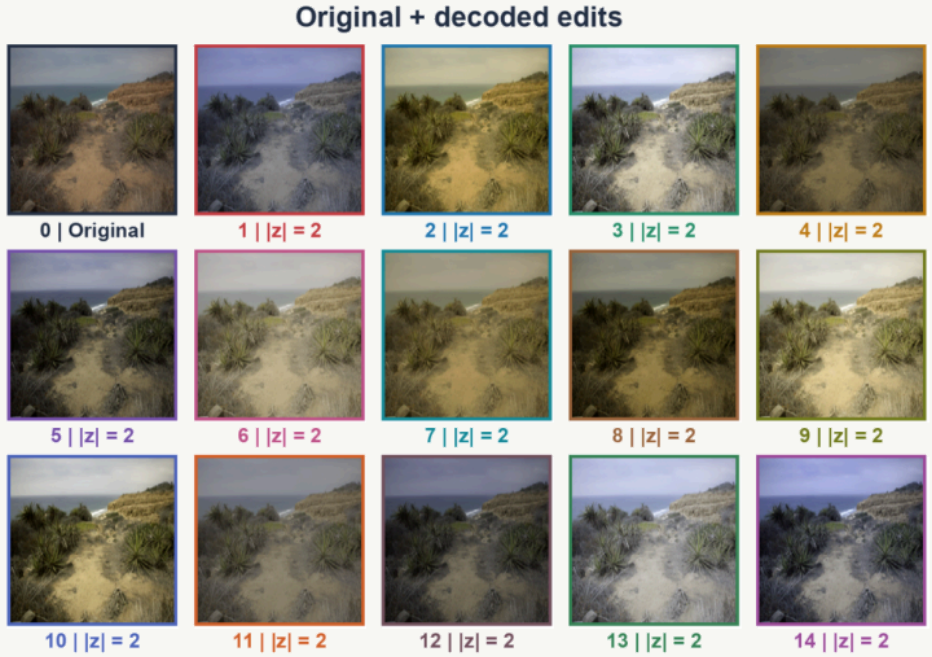
PPSD reconstruction PSNR

3D

Compact retouching space







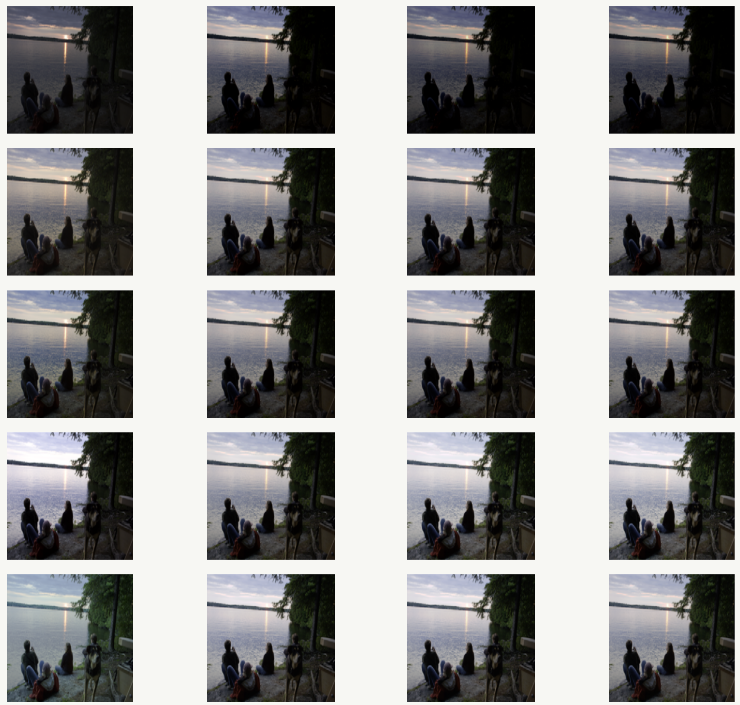
Preserves substantial global colour and tone information.

// 17 Quantitative results on FiveK



Same image · different expert histories

Selected training-image diagnostic · not test averages



FiveK PSNR (dB)

Expert E excluded from training

EXPERT	TOP-1	BEST-3*	BEST-5*
A	23.591	25.721	26.809
B	25.177	26.378	27.135
C	23.595	25.173	26.344
D	22.183	23.735	25.167
E	23.263	24.947	25.617
Mean	**23.562**	**25.191**	**26.214**

23.263 dB

Unseen Expert E · Top-1

0.8852

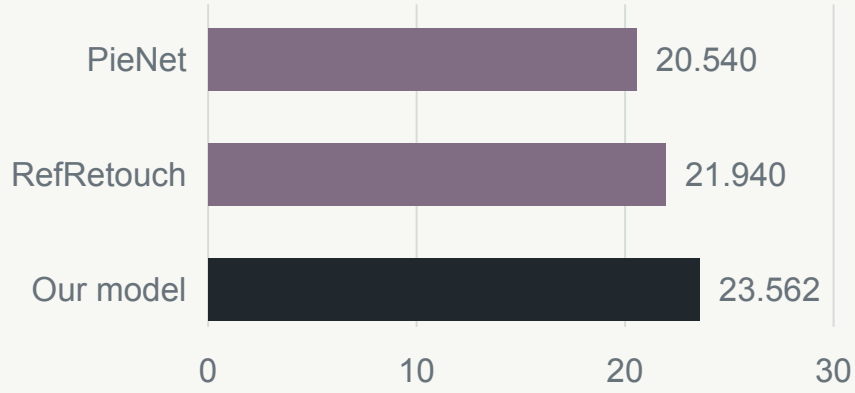
Mean Top-1 SSIM

***Best-of-K uses the target offline.** Higher scores show candidate coverage, not automatic best-edit selection.

Thin rays: sampled z vectors; thick rays: representatives; black: target. Direction density is normalised per image.

// 18 Comparison with existing methods

FiveK · Top-1 PSNR (dB)

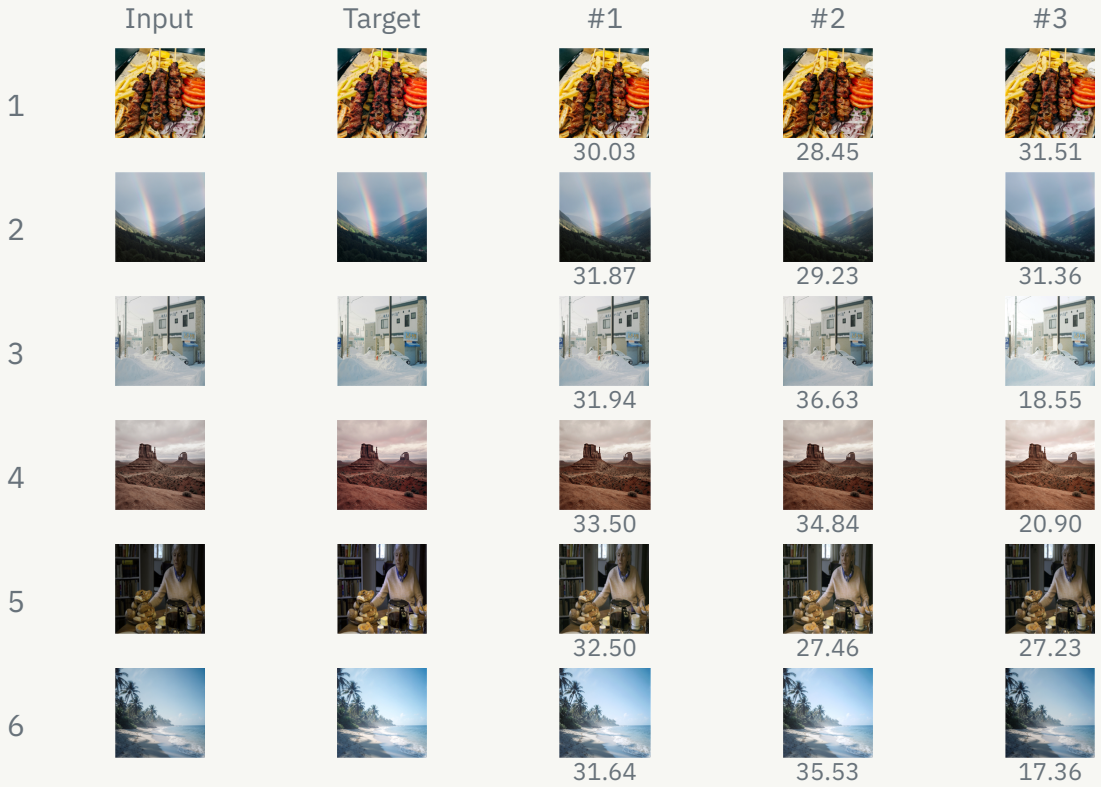


METHOD	SSIM	HISTORY / REFS
PieNet	**0.851**	**100**
RefRetouch	**0.9**	**100**
Our model	**0.8852**	**8**

PPSD: 24.5 dB PrefLUT: 22.094 dB · differing protocols

PPSD candidate examples

Selected examples from the supplied figure

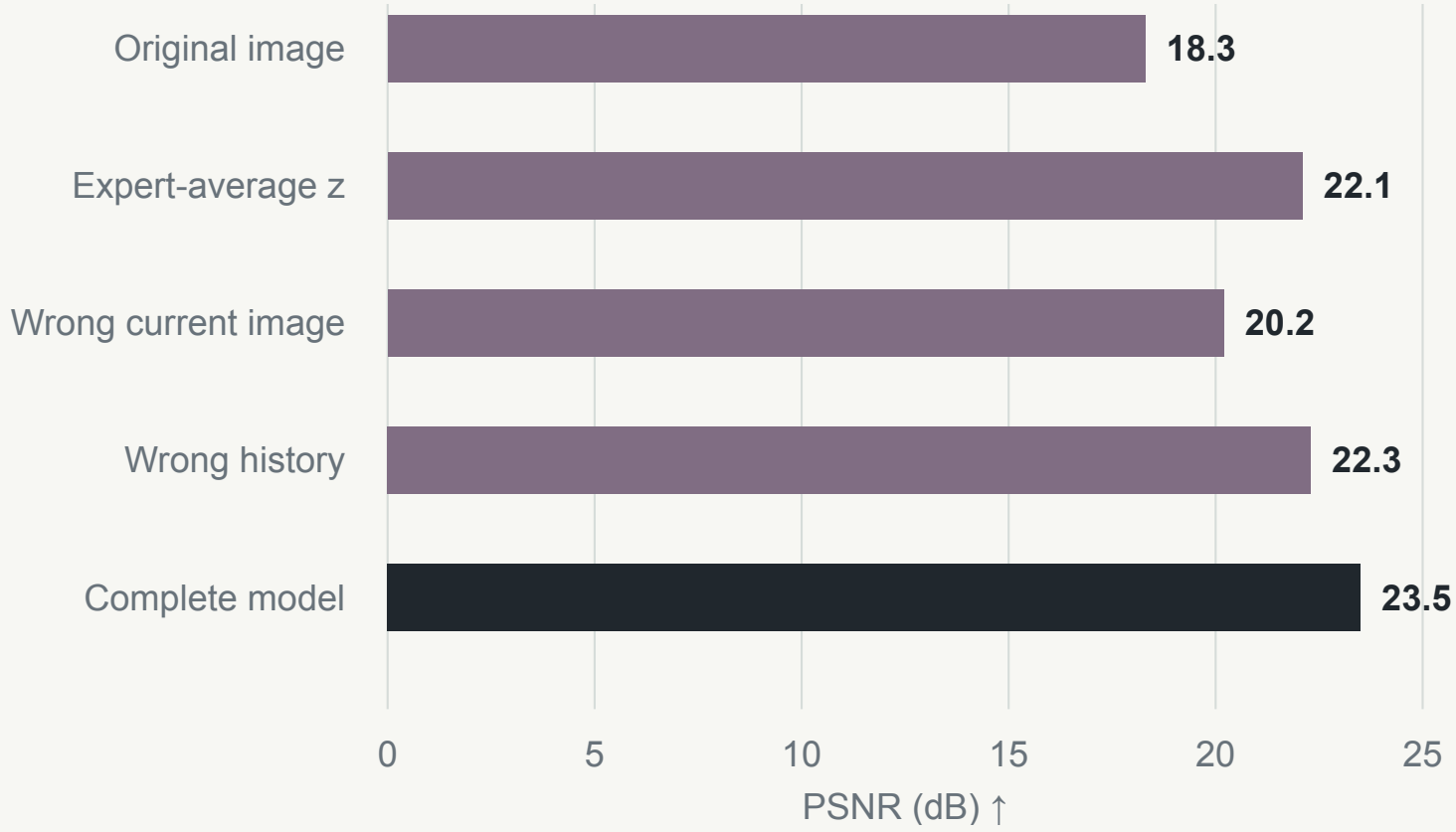


Candidate PSNR (dB) · model order; #1 need not score highest.

Performance context, not a matched benchmark. Splits and reference counts differ; RefRetouch has higher SSIM.

// 19 Ablation studies

Both conditions contribute. Current-image information has the larger effect.



−3.3 dB

Wrong current image

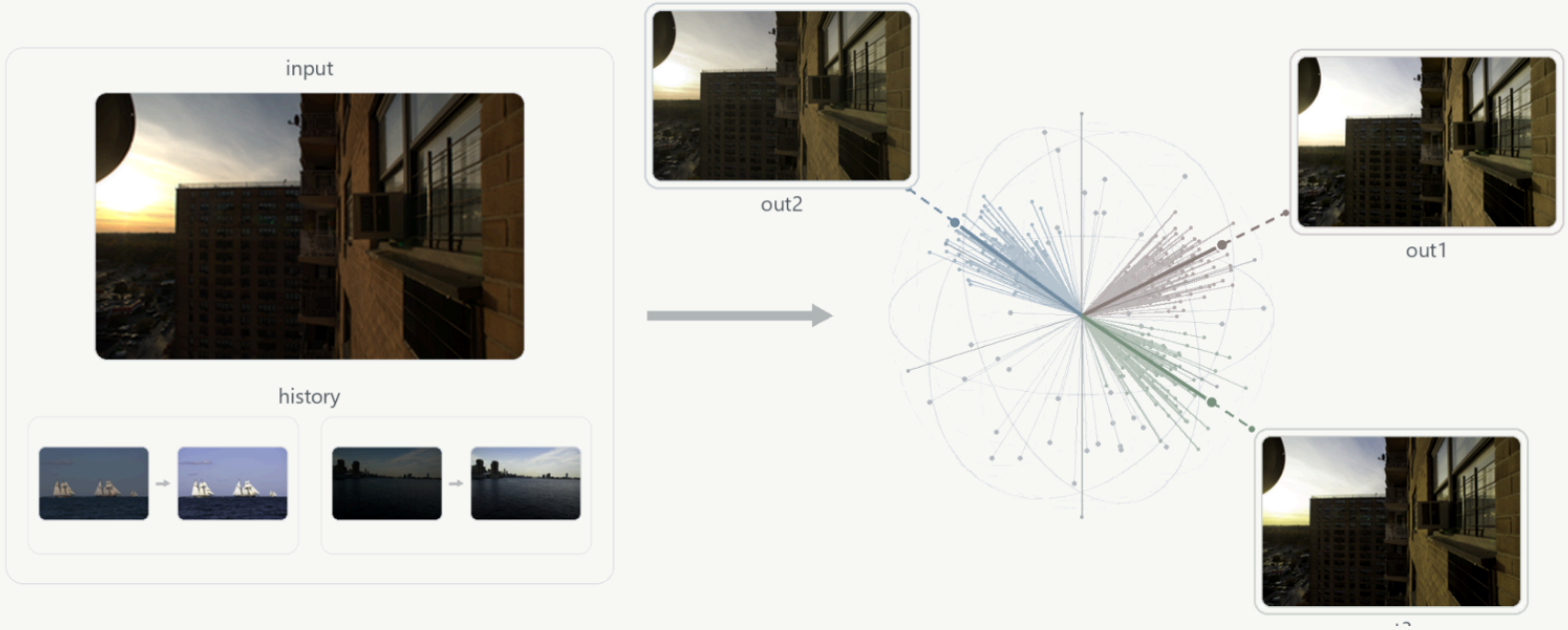
−1.2 dB

Wrong user history

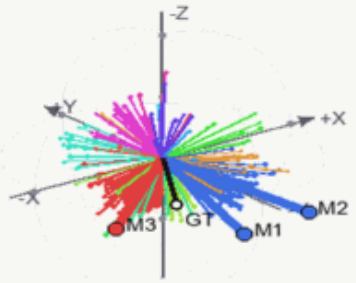
Relative to the complete model: 23.5 dB.

Current image + Historical preferences → **Conditional prediction**

// 20 Multimodal evaluation and test limitations



Conditioning → sampled edit space → multiple outputs



3D samples: direction + magnitude



Direction density: ignores magnitude

What the figures show

One condition can produce several candidate edits.

Thin rays: samples.
Marked rays: candidate vectors.
Black ray: recorded target.

Density radius shows local direction counts, normalised per image.

Testing limitations

Best-of-K uses ground truth offline.

One recorded edit cannot establish that every candidate is preferred.

Top: schematic with two history pairs drawn; the model uses eight. Bottom: a supplied PPSD result.

// 21 Discipline knowledge, implementation and tools

Python | PyTorch | CUDA

One integrated research pipeline

01 → CNN

Image features

Computer vision

02 → 3D z + LUTs

Compact edits

Representation learning

03 → Transformer

History context

Sequence learning

04 Flow

Conditional density

Probabilistic modelling

≈39.7 ms

Monte Carlo inference

115.9 MiB

Peak CUDA allocation

RTX 4060

Laptop GPU · 256 × 256

Batch 1, fresh history; inputs on GPU, historical z cached. Model-side timing excludes loading, disk I/O and application overhead.

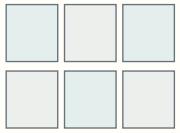
// 22 Client feedback and architecture evolution

From one edit to a distribution

Experiments + client feedback

01 →

Language + segmentation

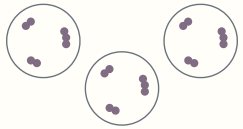


Instructions → regions

Segmentation errors; still needs instructions.

02 →

Dynamic grouping

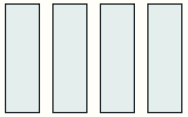


Similar edits → groups

Edit groups did not represent real users.

03 →

Memory slots



History → stored styles

Soft retrieval averaged edits; hard retrieval fixed styles.

04 →

Similarity retrieval

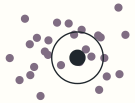


Image → nearest edit

≈30% accuracy; one neighbour was unreliable.

05 ✓

Conditional density

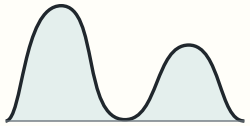


Image + history → p(z)

A distribution represents multiple possible edits.

Regular client discussions and experimental findings shaped the architecture.

// 23 Project management and collaboration

ENGINEERING 01

GitHub

Version control · integration

COMMUNICATION 02

Zoom + WeChat

Client reviews · team alignment

DELIVERY 03

9 members

Shared, evolving responsibilities

01 →

Research

Review literature

Agree on a testable hypothesis.

02 →

Prototype

Build & integrate

Coordinate model interfaces in GitHub.

03 →

Evaluate

Run experiments

Compare findings and feasibility.

04 ↺

Review

Discuss with client

Adjust priorities and responsibilities.

Self-managed research workflow with regular client communication.

// 24 Individual contributions

Nine contributors, one delivery chain.

YD Yuchen Deng

Prototype; integration; client coordination.

GX Gary Xu

Completed work not specified in Table 6.

SF Sihan Fang

Completed work not specified in Table 6.

YC Yiming Chen

Progress reflections; presentation and demo review.

YG Yiheng Guo

Doomsayer; deliverables not specified.

LW Lunyan Wang

Flow5 model; training; evaluation.

WS Weishao Su

Team management; meeting minutes.

XC XingTang Chen

Interactive slides; FiveK preparation; report.

ZZ Zihao Zhang

Tracker; deliverables not specified.

Contributions: report Table 6. Deliverable gaps remain explicit.

Research → development → evaluation → delivery

Reflections, limitations and conclusion

Personalization as a distribution of edits.

$P(z \mid \text{image}, \text{history})$



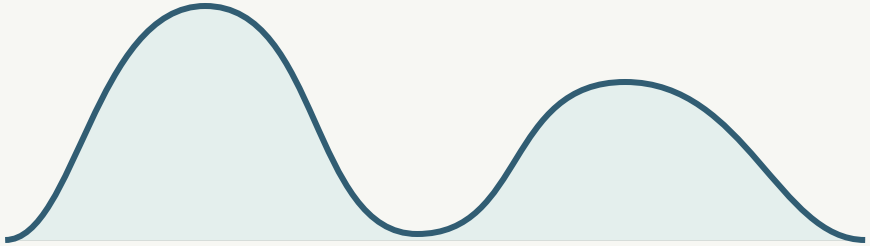
Original and recorded expert edit, FiveK a0412.

Current limitations

Weaker history dependence

Global colour and tone edits only

Limited subjective preference validation



Illustrative preference density

3D + LUTs
compact edits

Transformer
history context

Flow
multiple candidates

Thank you / Q&A